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**REPORT ON
PROJECT BASED LEARNING**

Carried out on

**Dermatology Disease Classification Using MLP
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CONTENT

No.	Title	Page No.
	Abstract	3
1	Introduction	3
2	Literature Review	4-5
3	Dataset Description	5-6
4	Methodology	6-7
5	Model Architecture	7
6	Results and Discussion	7-8
7	Conclusion	10-11
8	References	11-12

Abstract

Dermatological diseases are common medical conditions that require accurate and timely diagnosis for effective treatment. Manual diagnosis of skin diseases relies heavily on expert dermatologists and is often subjective due to overlapping clinical symptoms among different diseases. With the increasing availability of medical data, machine learning techniques have gained importance in developing automated diagnostic support systems.

This project presents an automated dermatological disease classification system using a Multi-Layer Perceptron (MLP) model. The proposed approach utilizes a dermatology dataset consisting of multiple clinical attributes describing various skin-related symptoms. Data preprocessing techniques such as handling missing values, feature scaling, and correlation analysis were applied to improve model learning and stability. The MLP model was trained and evaluated using appropriate classification metrics. Experimental results demonstrate that the proposed model achieves high classification accuracy with balanced precision and recall, indicating its effectiveness in assisting dermatologists in disease diagnosis.

Introduction

Skin diseases are among the most widespread health problems affecting individuals of all age groups across the world. Dermatological conditions can range from mild irritations to chronic and severe disorders that significantly affect a patient's quality of life. Accurate diagnosis is critical, as incorrect or delayed diagnosis may result in ineffective treatment and prolonged suffering.

Diagnosing skin diseases is challenging because many conditions exhibit similar clinical and visual characteristics such as erythema, scaling, itching, and inflammation. Traditional diagnostic methods rely primarily on clinical examination and the experience of dermatologists, which can be subjective and time-consuming. Additionally, access to experienced dermatologists may be limited in remote or underdeveloped regions.

With advancements in machine learning and artificial intelligence, automated decision-support systems have emerged as promising tools in healthcare. Machine learning models can analyze multiple clinical attributes simultaneously and identify complex patterns that may not be easily recognized through manual diagnosis. These systems can assist medical professionals by improving diagnostic accuracy, consistency, and efficiency.

The objective of this project is to design and implement an automated dermatological disease classification system using a Multi-Layer Perceptron (MLP) model. The system utilizes structured clinical data to classify patients into different dermatological disease categories. MLP is chosen due to its ability to learn non-linear relationships in tabular medical data. The proposed system aims to support dermatologists by providing reliable classification results and reducing diagnostic effort.

Literature Review

1) Xiangya-Derm — Classification of six common skin diseases (JMIR, 2021) { Accuracy $\approx$ 90.8%}

This study was focused on creating a large dermatology image dataset and used deep learning models to classify six common skin diseases. The researchers mainly used CNN models and achieved good classification performance. However, the approach required dermoscopic images, proper labeling, and expensive imaging equipment. This shows that image-based AI can be very effective, but our project focuses on clinical symptom data using an MLP model, which is simpler and easier to implement.

2) Skin Lesion Classification and Detection (MDPI Diagnostics, 2023) {95% accuracy range}

This study reviewed various AI techniques used for detecting skin diseases. It discussed different models, starting from traditional machine learning methods to advanced deep learning models such as CNNs and transfer learning models. We came to conclusion that deep learning models provide high accuracy but require large datasets and high computational resources. This highlights the importance of developing models that can work effectively with structured clinical data, as done in our project.

3) SSD-KD: Self-Supervised Diverse Knowledge Distillation (2022) {~85% accuracy on ISIC skin lesion classification dataset}

This research aimed to reduce the size and computational cost of deep learning models while maintaining good performance. The researchers used knowledge distillation techniques with CNN models to classify dermoscopic skin images. Although the model achieved good accuracy with reduced computation, it still depended on image data. In contrast, our project uses clinical symptom data, which makes the system more lightweight and practical.

Dataset Description

The dataset used in this project is a dermatology dataset containing structured clinical records of patients diagnosed with various skin diseases. Each instance in the dataset represents a single patient record.

The dataset consists of:

- **Number of instances:** 366
- **Number of input features:** 34
- **Type of data:** Tabular, numerical
- **Output type:** Multi-class categorical label

The 34 input features describe clinical symptoms and observations commonly used by dermatologists during diagnosis. These attributes include features related to erythema, scaling, papules, plaques, itching, and other dermatological indicators. Since these symptoms often co-occur in real medical conditions, some features exhibit correlation.

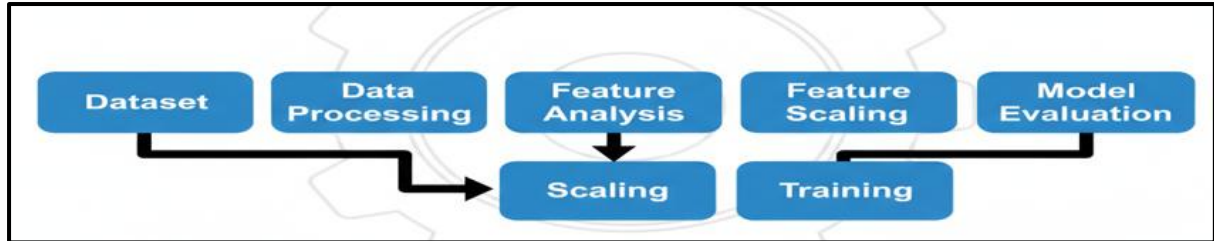
The target variable represents the diagnosed dermatological disease and is classified into six distinct classes:

1. **Psoriasis** – A chronic autoimmune skin disease characterized by red, scaly patches and inflammation.
2. **Seborrheic Dermatitis** – A condition causing scaly patches and red skin, commonly affecting areas rich in oil glands.
3. **Lichen Planus** – An inflammatory disease presenting as purplish, itchy, flat-topped lesions.
4. **Pityriasis Rosea** – A self-limiting skin rash that usually begins with a single large patch followed by smaller lesions.
5. **Chronic Dermatitis** – Long-term skin inflammation associated with itching, redness, and irritation.
6. **Pityriasis Rubra Pilaris** – A rare skin disorder characterized by reddish-orange scaling patches and thickened skin.

Since each patient belongs to exactly one disease category, the dataset represents a multi-class classification problem, making it suitable for neural network-based models such as MLP.

Methodology

The methodology followed in this project consists of a systematic sequence of steps to ensure reliable and reproducible results. The overall workflow of the proposed system is as follows:



4.1 Data Preprocessing

Initially, the dataset was analyzed to understand its structure, feature types, and presence of missing values. Any missing values in the dataset were handled using median imputation, which is robust to outliers and preserves the central tendency of the data. This step ensures data completeness and prevents errors during model training.

	erythema	scaling	definite_borders	itching	koebner_phenomenon	polygonal_papules	follicular_papules	oral_mucosal_involvement	knee_and_elbow_involvement	scalp_involvement	...	disappearance_granular_layer
0	2	2	0	3	0	0	0	0	0	1	0	...
1	3	3	3	2	1	0	0	0	0	1	1	...
2	2	1	2	3	1	3	0	0	3	0	0	...
3	2	2	2	0	0	0	0	0	0	3	2	...
4	2	3	2	2	2	2	2	0	2	0	0	...

5 rows x 35 columns

Figure 1: Data Preprocessing dermatological dataset features.

4.2 Correlation Analysis

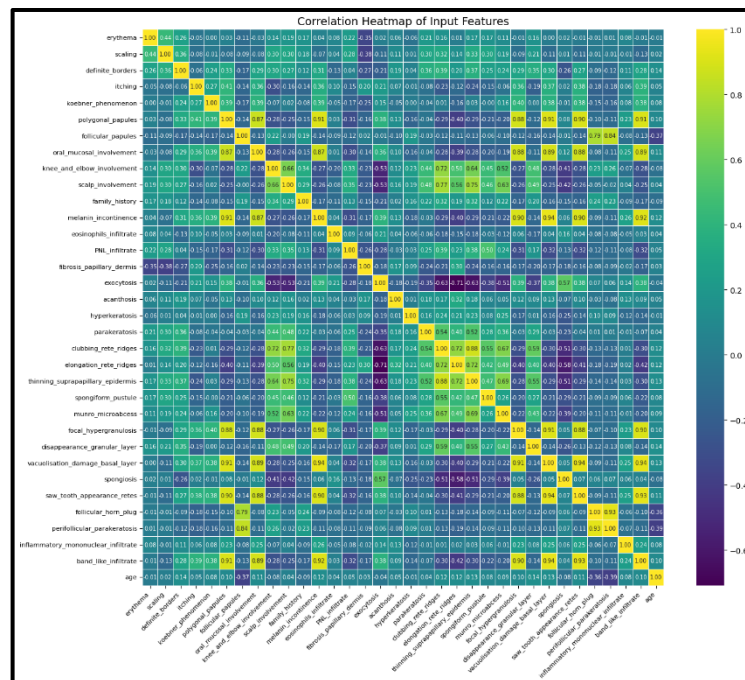


Figure 2: Correlation matrix of dermatological dataset features.

Explanation: -

A correlation matrix was computed to analyze the linear relationships among input features. Correlation analysis helps identify redundant features that convey similar information. Highly correlated features can lead to multicollinearity, which negatively affects model stability. A heatmap visualization of the correlation matrix was used to interpret feature dependencies visually.

4.3 Feature Scaling

Feature scaling was performed using standardization, where each feature was transformed to have zero mean and unit variance. Since MLP relies on gradient-based optimization, unscaled features can cause slow convergence or biased learning. Standardization ensures that all features contribute equally during training.

4.4 Train-Test Split

The dataset was split into training and testing subsets using an 80:20 ratio. The training set was used to learn model parameters, while the testing set was reserved for unbiased performance evaluation. This separation helps assess the generalization capability of the model.

4.5 Graphs and Visualizations

This project includes relevant visual representations to support data analysis and model evaluation.

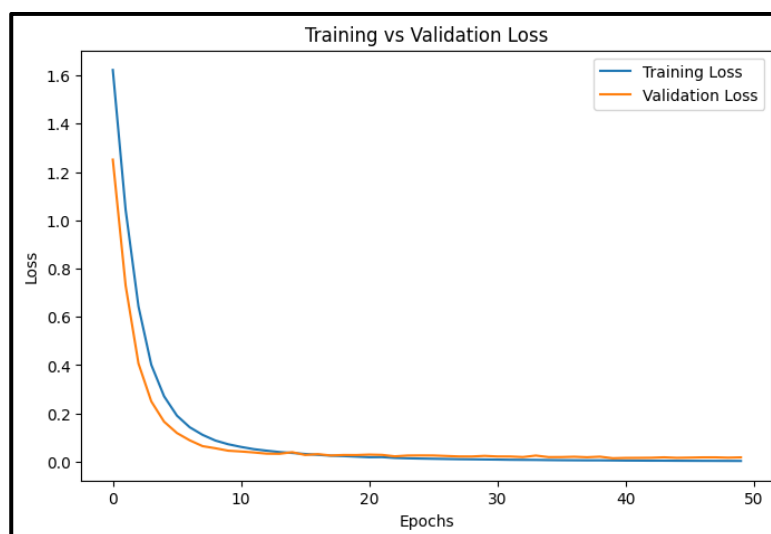


Figure 3: Training and validation loss curves of the MLP model.

Explanation: -

The loss curves show a steady decrease in both training and validation loss, indicating effective learning and convergence of the MLP model. The absence of a significant gap between the curves suggests minimal overfitting.

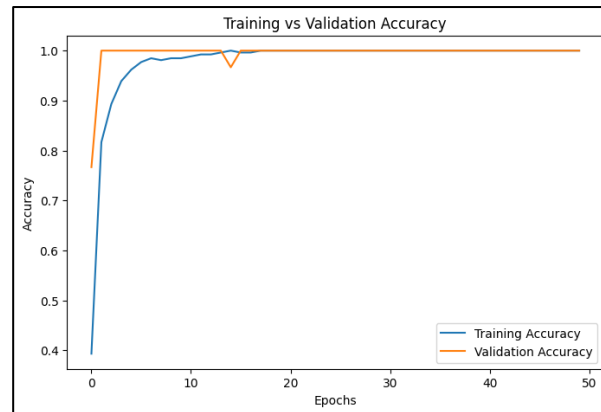


Figure 4: Training and validation accuracy curves of the MLP model

Explanation: -

The accuracy curves demonstrate rapid improvement during initial epochs, followed by stable high accuracy for both training and validation sets. This indicates good generalization performance of the MLP model.

Model Architecture

The Multi-Layer Perceptron model consists of the following components:

- **Input Layer:** Number of neurons equal to the number of input features
- **Hidden Layer 1:** 64 neurons with ReLU activation
- **Hidden Layer 2:** 32 neurons with ReLU activation
- **Output Layer:** Softmax activation for multi-class classification

ReLU activation is used to introduce non-linearity and avoid vanishing gradient problems. The softmax function ensures probabilistic interpretation of output classes.

Results and Discussion

The performance of the proposed MLP-based dermatological disease classification system was evaluated using multiple metrics to ensure comprehensive assessment.

6.1 Accuracy

The performance of the proposed Multi-Layer Perceptron model was evaluated on the test dataset using classification accuracy as the primary metric. The model achieved an overall test accuracy of 97.29%, indicating strong predictive capability in classifying dermatological disease categories.

6.2 Precision, Recall, and F1-Score

In addition to accuracy, the model performance was further evaluated using precision, recall, and F1-score to ensure balanced classification across all disease classes. The weighted precision of 0.977, recall of 0.973, and F1-score of 0.973 indicate that the model maintains a good balance between false positives and false negatives. These metrics are particularly important in medical diagnosis, where both misclassification and missed detection can have significant implications.

Metric	Value
Accuracy	97.29%
Precision	0.977
Recall	0.973
F1-score	0.973

Table : Performance Metrics of the Proposed MLP Model

6.2 Confusion Matrix Analysis

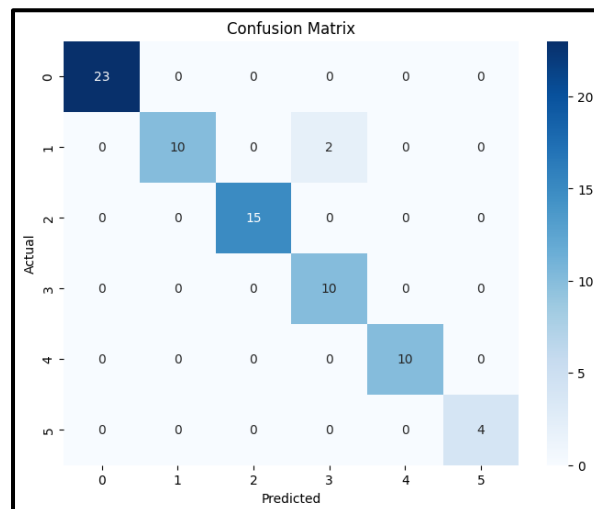


Figure 2: Confusion matrix of the MLP classifier on the test dataset.

Explanation: -

Figure 2 presents the confusion matrix obtained for the MLP classifier on the test dataset. The diagonal elements of the matrix represent correctly classified instances, while the off-diagonal elements indicate misclassifications.

Most samples are correctly classified, as evidenced by the strong diagonal values. Minor misclassifications occur between disease classes with similar clinical features, which is expected due to overlapping symptom characteristics in dermatological conditions.

Conclusion

This project successfully demonstrates the application of a Multi-Layer Perceptron (MLP) model for the classification of dermatological diseases using clinical data. The proposed system effectively captures the non-linear relationships between dermatological features and disease classes, resulting in high classification accuracy. Careful data preprocessing, correlation analysis, and feature scaling played a crucial role in improving the learning stability and performance of the MLP model.

Despite the encouraging results, the performance of the system is influenced by the quality and size of the dataset. Certain disease classes with similar clinical characteristics occasionally lead to misclassification, highlighting the inherent complexity of dermatological diagnosis.

Future improvements to this work include experimenting with alternative machine learning models such as Random Forest and Support Vector Machines for comparative analysis. Further performance gains may also be achieved through systematic hyperparameter tuning and by training the model on larger and more diverse datasets. These enhancements can improve generalization and make the system more robust for real-world clinical applications.

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